Assignment-Regression Algorithmn

Problem Statement:

Answer 1: Predicting the insurance charges based on the parameters

- Input is number, so domain is learning.
- Requirements are clear, input and output fields are clear, hence it is supervised learning
- Output values are continuous numerical values, so it is regression.

Answer 2: The Basic information for this dataset is Age, Sex, BMI, Children, Smoker and the output Insurance charges. Total number of rows are 1338 rows and 6 columns (5 input and 1 output) from the given dataset.

Answer 3: Need to convert the Sex (male/female) and Smoker columns (yes/no) input fields as it is required to calculate the insurance charges. Both these data are Nominal as it is not comparable and it is like noun and we need to go for One Hot encoding.

Answer 4: Created Multiple Linear regression, Support Vector machine, Decision Tree Regressor and Random Forest Regressor models and deployed the finalized best model which is Random Forest Regressor. Below are the r2 results of each models.

1. Multiple Linear Regression
Combinations of Parameters

S.NO	fit_intercept	copy_X	tol	n_jobs	positive	r2_score
1	TRUE	TRUE	1.00E-06	None	FALSE	0.789479035
2	FALSE	TRUE	1.00E-06	None	FALSE	0.762372328
3	FALSE	FALSE	1.00E-06	None	FALSE	0.762372328
4	TRUE	TRUE	1.00E-06	1000	FALSE	0.789479035
5	TRUE	TRUE	10	1000	FALSE	0.789479035
6	TRUE	TRUE	10000	None	FALSE	0.789479035
7	TRUE	TRUE	1.00E-06	None	TRUE	0.789442939

2.Support Vector Machine regression
Combinations of Parameters

S.NO	kernel	gamma	r2_score
1	'rbf'	'scale'	-0.08843
2	poly'	'scale'	-0.0757
3	linear	'scale'	-0.0101
4	sigmoid	scale'	-0.07543
5	'rbf'	auto'	-0.08338
6	linear	auto'	-0.0101
7	poly'	auto'	-0.0757
8	sigmoid	auto'	-0.07543

3.**DecisionTreeRegressor**

Combination of Parameters

S.NO	criterion	splitter	max_features	r2_score
1	squared_error	best	None	0.692160791
2	friedman_mse	best	None	0.708935902
3	absolute_error	best	None	0.66779936
4	poisson	best	None	0.718978267
5	squared_error	random	None	0.741721661
6	friedman_mse	random	None	0.728890259
7	absolute_error	random	None	0.750998014
8	poisson	random	None	0.731111167
9	squared_error	best	sqrt	0.716531476
10	friedman_mse	best	sqrt	0.693866537
11	absolute_error	best	sqrt	0.666407737
12	poisson	best	sqrt	0.728693055
13	squared_error	random	sqrt	0.628978417
14	friedman_mse	random	sqrt	0.709891819
15	absolute_error	random	sqrt	0.660394846
16	poisson	random	sqrt	0.700811034
17	squared_error	best	log2	0.735214112
18	friedman_mse	best	log2	0.727602454
19	absolute_error	best	log2	0.721249315
20	poisson	best	log2	0.705666112
21	squared_error	random	log2	0.510625521
22	friedman_mse	random	log2	0.669135688
23	absolute_error	random	log2	0.736434537
24	poisson	random	log2	0.64752601

4. Random Forest Regressor

Combination of Parameters

S.NO	n_estimators	criterion	max_features	random_state	r2_score
1	100	squared_error	None	None	0.855118895
2	100	friedman_mse	None	None	0.853390996
3	100	absolute_error	None	None	0.851914384
4	100	poisson	None	None	0.853454758
5	50	squared_error	None	None	0.849796265
6	50	friedman_mse	None	None	0.854321973
7	50	absolute_error	None	None	0.854163779
8	50	poisson	None	None	0.854616333
9	200	squared_error	sqrt	None	0.871912908
10	200	friedman_mse	sqrt	None	0.871608492
11	200	absolute_error	sqrt	None	0.872247861
12	200	poisson	sqrt	None	0.869395834

13	400	squared_error	log2	None	0.872317178
14	400	friedman_mse	log2	None	0.87221664
15	400	absolute_error	log2	None	0.874148632
16	400	poisson	log2	None	0.871562271

Answer 6:

Out of these 4 models, finalizing the model would be Random Forest Regressor as it is maximum value nearer to 1 which is `0.874148632 (highlighted in yellow)`, other models are having less values when comparing to this model value.